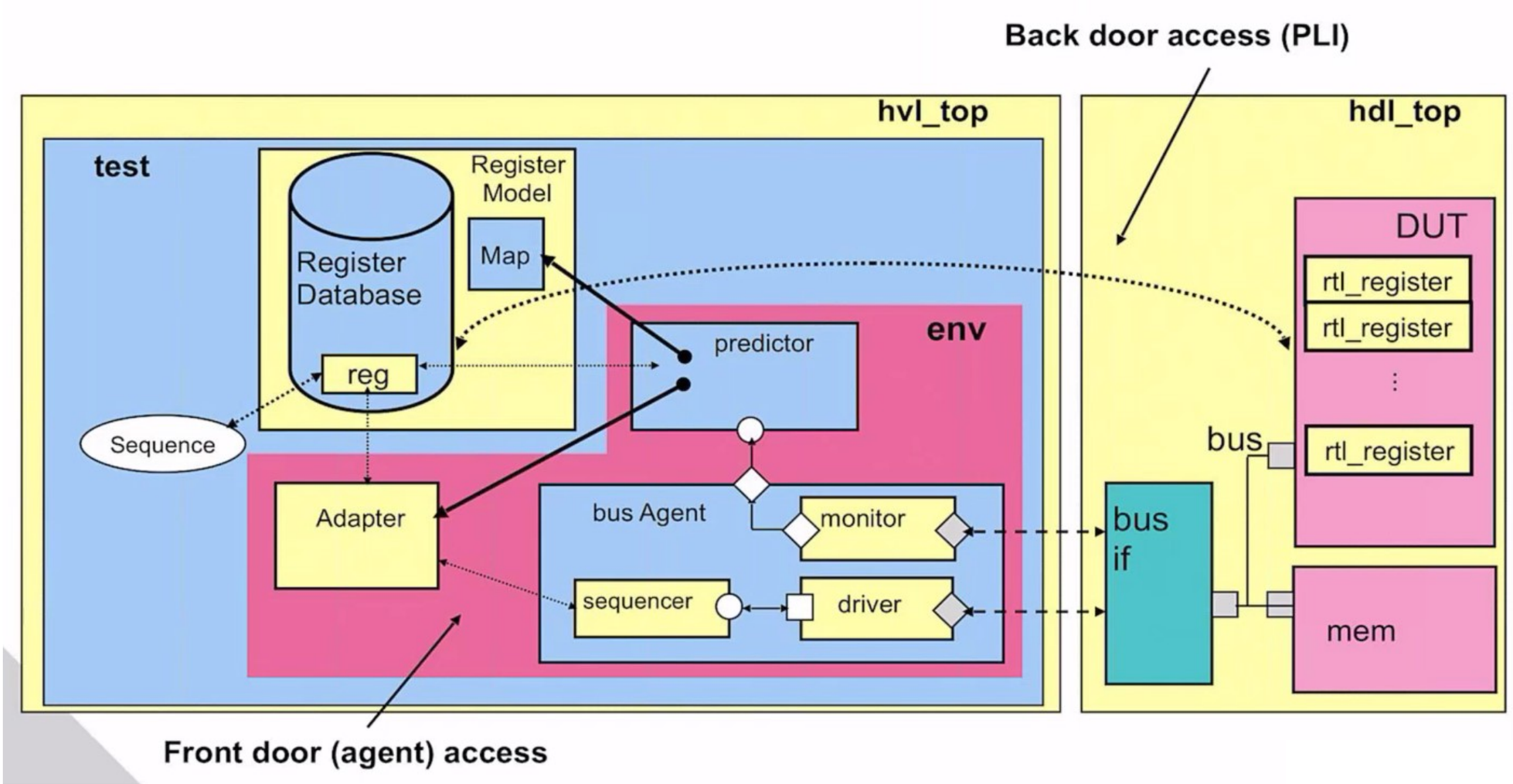
Test Case ID	Test Case Name	Description	UVM Sequence(s) Used	Key Features Verified	Stimulus Details	Expected Outcome	Priority	Status
REG_ACCESS								
Pattern test	Comprehensive RAL Access	Verifies R/W via FD & BD for all registers and fields using various data patterns. Checks model synchronization.	`RegisterFile_reg_seq`	REG_RW, REG_BD, REG_FD, REG_DATA, REG_RAL_SYNC	All registers; patterns (0, all 1s, walking, random); FD/BD R/W;	All RAL operations (write/read/poke/peek/field ops) complete with UVM_IS_OK. Read/peeked data matches written/poked data. Mirrored values are correct.	High	Implemented
STRESS_TESTING								
Randomized test	RAL Random Stress Test	Subjects registers to a high volume of randomized read/write operations via FD.	Custom random sequence in `RegisterFile_reg_seq`	REG_RW, REG_DATA, REG_RAL_SYNC	N (large number) random FD R/W doperations to random valid addresses with random data.	All operations complete with UVM_IS_OK. Data integrity maintained. Model synchronized. No unexpected DUT behavior.	High	Implemented
FIELD_TESTING								
ss_reg field	RAL Field Test	Write/Poke JUST the char_len field of ss_reg	Test case 4 (label) in `RegisterFile_reg_seq`	REG_FIELD	Change character field of ss_reg	Only the char_len field should be modified	Low	Implemented


Back door access (PLI)
hvl_top
hdl_top
test
Register Model
Register Database
Map
reg
predictor
env
Sequence
Adapter
bus Agent
monitor
sequencer
driver
DUT
rtl_register
rtl_register
rtl_register
bus
bus if
mem
Front door (agent) access